Biochemistry Cume (Protein/peptide Chemistry)

October 29, 2010

Total points:100

1. A protein was subjected to end group analysis by dansyl chloride. The liberated amino acids are found to be present with a molar ratio of two parts **Ala** to one part **Leu**. What conclusions can be drawn concerning the structure of the protein? (**10 points**)
2. What are the cleavage products formed when the polypeptide chains (given below) are subjected to different treatments (as indicated in parentheses)
   - **a.** Ser-Ala-Phe-Lys-Pro (subjected to chymotrypsin treatment)
   - **b.** Thr-Cys-Gly-Met-Asn (subjected to cynogen bromide treatment)
   - **c.** Val-Trp-Lys-Pro-Arg-Glu (subjected to trypsin treatment) (**30 points**)
3. A polypeptide was subjected to the following treatments with indicated results. What is the primary structure of the polypeptide
   - A. Acid hydrolysis of the polypeptide yielded,
     (Ala, Arg, Cys, Glx, Gly, Lys, Leu, Met,Phe, Thr)
   - B. Aminopeptidase treatment yielded no fragments
   - C. Carboxypeptidase treatment yielded no fragments
   - D. Trypsin treatment of the polypeptide followed by Edman degradation yielded the following products: Cys-Gly-Leu-Phe-Arg and Thr-Ala-Met-Glu-Lys (**60 points**)

   <u>For full credit on question 3, the rationale for the results obtained after each treatment must be provided.</u>